

XXX PHARMACEUTICALS, INC.
Clinical Development

CLINICAL STUDY PROTOCOL

Protocol Number: HTN101-301

A Phase III, Randomized, Double-Blind, Placebo-Controlled, Parallel-Group, Multicentre Study to Evaluate the Efficacy and Safety of HTN-101 50 mg Once Daily in Adult Subjects with Essential Hypertension

Investigational Product	HTN-101 (film-coated tablet, 50 mg) and matching placebo
Indication	Essential Hypertension
Phase	III
Sponsor	XXX Pharmaceuticals, Inc.
Protocol Version	Version 2.0 (Amendment 1)
Version Date	15 January 2024
NCT Number	Not applicable — synthetic training protocol

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Prepared as part of a statistical programming portfolio project. This protocol, the sponsor, investigators, sites, and all subject data are fictional and were created solely to demonstrate CDISC SDTM/Adam and clinical statistical programming workflow.

PROTOCOL SYNOPSIS

Title	A Phase III, Randomized, Double-Blind, Placebo-Controlled, Parallel-Group, Multicentre Study to Evaluate the Efficacy and Safety of HTN-101 50 mg Once Daily in Adult Subjects with Essential Hypertension
Protocol Number	HTN101-301
Phase	III
Sponsor	XXX Pharmaceuticals, Inc.
Study Design	Randomized, double-blind, placebo-controlled, parallel-group, multicentre study
Study Population	Adult men and women, 40 to 79 years of age, with essential hypertension
Total Planned Subject	100
Treatment	50 HTN-101 / 50 Placebo
Number of Sites	Approximately 6 investigational sites
Planned Sample Size	Approximately 50 subjects randomized 1:1 (25 per treatment arm)
Treatment Duration	12 weeks (84 days) double-blind treatment
Investigational Product & Dose	HTN-101 50 mg, oral tablet, once daily; matching placebo once daily
Primary Objective	To evaluate the efficacy of HTN-101 50 mg once daily compared with placebo in reducing seated systolic blood pressure (SBP) from baseline to Week 12
Primary Endpoint	Change from baseline in seated SBP at Week 12
Secondary Objective	To evaluate the efficacy of HTN-101 50 mg once daily compared with placebo in reducing seated diastolic blood pressure (DBP) from baseline to Week 12
Secondary Endpoint	Change from baseline in seated DBP at Week 12
Safety Objective	To evaluate the safety and tolerability of HTN-101 50 mg once daily compared with placebo
Statistical Methods	Primary and secondary endpoints analysed using a mixed model for repeated measures (MMRM), with treatment, visit, treatment-by-visit interaction, and baseline value as model terms. Safety data summarized descriptively.

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LIST OF ABBREVIATIONS

Abbreviation	Definition
Adam	Analysis Data Model
AE	Adverse Event
ANCOVA	Analysis of Covariance
BP	Blood Pressure
CDISC	Clinical Data Interchange Standards Consortium
CRF	Case Report Form
DBP	Diastolic Blood Pressure
ET	Early Termination
GCP	Good Clinical Practice
ICH	International Council for Harmonisation
ITT	Intent-to-Treat
IRB/IEC	Institutional Review Board / Independent Ethics Committee
IWRS	Interactive Web Response System
MMRM	Mixed Model for Repeated Measures
PP	Per-Protocol
QD	Once Daily
SAE	Serious Adverse Event
SAP	Statistical Analysis Plan
SBP	Systolic Blood Pressure
SDTM	Study Data Tabulation Model
TEAE	Treatment-Emergent Adverse Event
TLF	Tables, Listings, and Figures
ULN	Upper Limit of Normal

1. INTRODUCTION

1.1 Background

Essential hypertension remains one of the most prevalent, modifiable risk factors for cardiovascular morbidity and mortality worldwide. Despite the availability of multiple approved antihypertensive drug classes, a substantial proportion of treated patients fail to achieve guideline-recommended blood pressure targets, and residual cardiovascular risk persists even among patients on established therapy.

HTN-101 is an investigational oral antihypertensive agent under development by XXX Pharmaceuticals, Inc. for the treatment of essential hypertension. Nonclinical and Phase I/II data (on file with the Sponsor) support further evaluation of the 50 mg once-daily dose in a Phase III confirmatory setting.

1.2 Rationale for the Study

This Phase III study is designed to confirm the blood-pressure-lowering efficacy of HTN-101 50 mg once daily relative to placebo over a 12-week treatment period, and to further characterize its safety and tolerability profile in a broader adult population with essential hypertension.

2. STUDY OBJECTIVES AND ENDPOINTS

2.1 Primary Objective and Endpoint

Objective: To evaluate the efficacy of HTN-101 50 mg once daily compared with placebo in reducing seated systolic blood pressure (SBP) from baseline to Week 12 in adults with essential hypertension.

Endpoint: Change from baseline in seated SBP at Week 12 ($\text{CHG} = \text{Week 12 SBP} - \text{Baseline SBP}$).

2.2 Secondary Objective and Endpoint

Objective: To evaluate the efficacy of HTN-101 50 mg once daily compared with placebo in reducing seated diastolic blood pressure (DBP) from baseline to Week 12.

Endpoint: Change from baseline in seated DBP at Week 12.

2.3 Safety Objective

Objective: To evaluate the safety and tolerability of HTN-101 50 mg once daily compared with placebo, as assessed by treatment-emergent adverse events (TEAEs), vital signs, and clinical laboratory parameters.

3. INVESTIGATIONAL PLAN

3.1 Overall Study Design

This is a randomized, double-blind, placebo-controlled, parallel-group, multicentre study conducted at approximately 6 sites. Eligible subjects will be randomized in a 1:1 ratio to receive HTN-101 50 mg once daily or matching placebo for 12 weeks.

3.2 Visit Schedule

Visit	Study Day	Purpose
Screening	Day –14	Eligibility assessment
Baseline / Randomization	Day 0	Baseline measurements; randomization
Week 4	Day 28	Treatment assessment
Week 8	Day 56	Treatment assessment
Week 12 / End of Treatment	Day 84	Primary and secondary endpoint assessment
Early Termination	Variable	Completed by subjects discontinuing early
Unscheduled	Variable	Performed at investigator discretion

3.3 Number of Subjects

Approximately 50 subjects will be randomized (25 per treatment arm) across approximately 6 investigational sites, allowing for an anticipated early-discontinuation rate of approximately 20%.

3.4 End of Study

End of study is defined as the date of the last visit or contact for the last subject participating in the study.

4. STUDY POPULATION

4.1 Inclusion Criteria

1. Male or female, 40 to 79 years of age, inclusive, at screening.
2. Documented diagnosis of essential hypertension.
3. Mean seated SBP 140–179 mmHg and mean seated DBP 90–109 mmHg at both screening and baseline.
4. Body mass index (BMI) 18–40 kg/m² at screening.
5. Willing and able to provide written informed consent prior to any study-related procedures.
6. Willing and able to comply with study visits and procedures for the duration of the study.
7. Women of childbearing potential must have a negative pregnancy test at screening and agree to use a medically acceptable method of contraception throughout the study.

4.2 Exclusion Criteria

1. Secondary hypertension of any etiology.
2. Mean seated SBP $\geq$ 180 mmHg or DBP $\geq$ 110 mmHg at screening.
3. History of myocardial infarction, stroke, or hospitalization for unstable angina within 6 months prior to screening.
4. Type 1 diabetes mellitus or poorly controlled Type 2 diabetes mellitus (HbA1c $>$ 9%).
5. Estimated glomerular filtration rate $<$ 45 mL/min/1.73m² or clinically significant hepatic impairment.
6. Known hypersensitivity to HTN-101 or its excipients.
7. Use of prohibited concomitant antihypertensive therapy that cannot be discontinued or held stable per protocol.

8. Pregnant or breastfeeding women.
9. Participation in another investigational drug or device study within 30 days prior to screening.
10. Any condition that, in the investigator's judgment, would compromise subject safety or the integrity of the study data.

5. STUDY TREATMENTS

5.1 Investigational Product

HTN-101 is supplied as a 50 mg film-coated tablet. Matching placebo tablets are identical in appearance, packaging, and labelling to maintain the blind.

5.2 Dosage and Administration

Subjects will self-administer one tablet (HTN-101 50 mg or matching placebo) orally, once daily in the morning, with or without food, for 12 weeks.

5.3 Randomization and Blinding

Eligible subjects will be randomized 1:1 to HTN-101 50 mg or placebo via an Interactive Web Response System (IWRS), stratified by site. Study drug, sponsor personnel involved in data analysis, investigators, site staff, and subjects will remain blinded to treatment assignment until database lock, except in cases of medical emergency requiring unblinding.

5.4 Treatment Compliance

Compliance will be assessed at each visit by tablet count and review of the subject's dosing diary. Subjects with compliance outside 80%–120% will be discussed with the medical monitor.

5.5 Prior and Concomitant Medications

Use of non-study antihypertensive medications is prohibited during the double-blind treatment period unless required for safety. All concomitant medications will be recorded from screening through the end-of-study visit.

6. STUDY PROCEDURES AND ASSESSMENTS

6.1 Schedule of Assessments

Assessment	Screening	Baseline	Wk. 4	Wk. 8	Wk. 12/ET
Informed consent	X	—	—	—	—
Demographics	X	—	—	—	—
Medical history	X	—	—	—	—
Inclusion / exclusion review	X	X	—	—	—
Physical examination	X	X	—	—	X
Vital signs (SBP/DBP/HR)	X	X	X	X	X
Body weight	X	X	X	X	X
12-lead ECG	X	X	—	—	X
Clinical laboratory tests	—	X	X	X	X
Pregnancy test (if applicable)	X	—	—	—	—
Concomitant medication review	X	X	X	X	X
Adverse event assessment	—	X	X	X	X
Study drug dispensing / accountability	—	X	X	X	X
Randomization	—	X	—	—	—

6.2 Efficacy Assessments

Blood pressure will be measured in triplicate in the seated position after a minimum 5-minute rest period, using a validated automated Oscillo metric device. The mean of the three measurements will be used as the SBP/DBP value for that visit for analysis purposes.

6.3 Safety Assessments

Safety will be assessed throughout the study via adverse event monitoring, vital signs, physical examination, clinical laboratory tests (including serum potassium and creatinine), and 12-lead ECG as specified in the Schedule of Assessments.

7. ADVERSE EVENT REPORTING

An adverse event (AE) is any untoward medical occurrence in a subject administered study drug, regardless of causal relationship. A serious adverse event (SAE) is any AE that results in death, is life-threatening, requires or prolongs hospitalization, results in persistent or significant disability, is a congenital anomaly, or is otherwise considered medically important.

All AEs will be recorded from the time of informed consent through the end-of-study/early-termination visit, including onset date, resolution date (or ongoing status), severity (mild, moderate, severe), seriousness, causality to study drug, action taken, and outcome. SAEs must be reported to the Sponsor or designee within 24 hours of investigator awareness.

8. STATISTICAL METHODS

8.1 Sample Size Determination

Assuming a between-group difference in mean change from baseline SBP of 12 mmHg, a common standard deviation of 15 mmHg, a two-sided significance level of 0.05, and a two-sample t-test, 25 evaluable subjects per treatment arm (50 total) provide approximately 80% power to detect the assumed treatment difference. This sample size is considered appropriate for the exploratory scope of this exercise; a fully powered registrational study would typically enrol a substantially larger population.

8.2 Analysis Populations

- Intent-to-Treat (ITT) / Full Analysis Set: All randomized subjects who receive at least one dose of study drug, analysed according to randomized treatment.
- Per-Protocol (PP) Population: ITT subjects without major protocol deviations affecting the primary endpoint.
- Safety Population: All subjects who receive at least one dose of study drug, analysed according to actual treatment received.

8.3 Primary Efficacy Analysis

Change from baseline in SBP at each post-baseline visit will be analysed using a mixed model for repeated measures (MMRM) with fixed effects for treatment, visit, treatment-by-visit interaction, and baseline SBP, and an unstructured covariance to model within-subject correlation. The treatment comparison at Week 12 will be based on the least-squares mean difference between HTN-101 and placebo.

8.4 Secondary Efficacy Analysis

Change from baseline in DBP at Week 12 will be analysed using the same MMRM approach specified for the primary endpoint. The secondary endpoint will be tested only if the primary endpoint achieves statistical significance, following a fixed-sequence multiplicity control procedure.

8.5 Safety Analyses

Treatment-emergent adverse events (TEAEs) will be summarized by System Organ Class and Preferred Term, treatment group, severity, and relationship to study drug for the Safety Population. Vital signs and clinical laboratory results will be summarized descriptively by visit and treatment group, including shift from baseline relative to reference ranges.

8.6 Handling of Missing Data

The MMRM approach for efficacy analyses accommodates missing data under a missing-at-random (MAR) assumption without explicit imputation. Sensitivity analyses (e.g., multiple imputation under a missing-not-at-random scenario) may be conducted to assess the robustness of the primary conclusion.

8.7 Interim Analysis

No formal interim analysis is planned for this study.

9. DATA MANAGEMENT AND QUALITY CONTROL

Data will be collected via an electronic case report form (eCRF) and mapped to CDISC SDTM domains, followed by derivation of CDISC Adam analysis datasets. All statistical outputs (tables, listings, and figures) will be produced from validated Adam datasets in accordance with the Statistical Analysis Plan (SAP).

Independent double programming and QC review will be performed for the primary efficacy tables and for all datasets supporting key safety and efficacy conclusions prior to database lock and finalization of the Clinical Study Report.

10. ETHICAL CONSIDERATIONS

This study will be conducted in accordance with the ethical principles originating from the Declaration of Helsinki and consistent with ICH E6(R2) Good Clinical Practice (GCP) guidelines and applicable regulatory requirements. The protocol, informed consent form, and any amendments will be reviewed and approved by an Institutional Review Board / Independent Ethics Committee (IRB/IEC) prior to study initiation at each site. Written informed consent will be obtained from each subject prior to any study-related procedures.

11. REFERENCES

1. International Council for Harmonisation. ICH E6(R2): Guideline for Good Clinical Practice.
2. International Council for Harmonisation. ICH E9: Statistical Principles for Clinical Trials.
3. Clinical Data Interchange Standards Consortium (CDISC). Study Data Tabulation Model Implementation Guide (SDTMIG).
4. Clinical Data Interchange Standards Consortium (CDISC). Analysis Data Model Implementation Guide (ADaMIG).

SIGNATURE PAGE

By signing below, the Investigator confirms agreement to conduct the study in accordance with this protocol, applicable regulatory requirements, and ICH GCP guidelines.

Investigator Signature: _____ Date: _____

Printed Name: _____

Site: _____